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ORIGINAL ARTICLE

Epidemic profile of common respiratory viruses in association SARS CoV-2 among SARI and ARI-two year study

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Abstract

Background COVID-19/SARS CoV-2 continue to pose a threat to human health and placed millions of livelihoods at risk. Surveillance for the other circulating seasonal viruses during this pandemic is necessary to understand the manifestations of the CoV-2 pandemic and their incidence.

Methods A detailed study survey was performed on subjects with acute respiratory infections (ARI) and severe acute respiratory infection (SARI) in the King Institute of Preventive Medicine and Research, Chennai from April 2020 to March 2022. A total of 1480 patients presenting with either SARI (41.8%) or ARI (58.1%) were screened for SARS CoV-2 and other respiratory viruses. The SARS CoV-2 real-time PCR was carried out using ICMR-approved kits and other respiratory viruses were detected using the commercially available real-time kit.

Results Out of the 620 SARI patients, 198 (31.9%) were positive for SARS CoV-2 RNA. Out of the 860 ARI patients, 352 (40.9%) were positive for SARS CoV-2 RNA. Among the 550 patients positive for SARS CoV-2, 7 (1.2%) were positive coexistent with other respiratory viruses. Among the 930 patients with negative SARS CoV-2, 222 (23.8%) were positive for other common respiratory viruses ($p=0.001$). Influenza viruses (36.9%) predominated followed by RSV (31.9%) and Parainfluenza virus (13.5%).

Conclusion This study suggests that viral coinfections are significantly higher among SARS CoV-2 negative individuals (23.8 vs. 1.2%). It is possibly due to viral interference and the competitive advantage of SARS CoV-2 in modulating the host immunity. Continuous surveillance is necessary for understanding the viral co-infection mechanisms.

Keywords Respiratory viruses · Influenza virus · Co-infections · SARS CoV-2 · SARI · ARI · COVID-19

Introduction

The Coronavirus disease 2019 (COVID-19), also known as the severe acute respiratory syndrome coronavirus 2 (SARS CoV-2), was first identified in China in December 2019. This novel virus has become pandemic and diagnosing SARS CoV-2 has become pivotal to early introduction of operational infection control measures and appropriate timely antiviral agents [1]. It is well known that SARS CoV-2 spreads through close contact, respiratory droplets/aerosols from an infected person and contaminated environment very similar to the already circulating influenza and other respiratory viruses. The COVID-19 has affected worldwide and had modified the dynamics of occurrence of other respiratory viral infections globally [2]. The emergence of COVID-19 co-infections with other respiratory pathogens poses a new concern as it can complicate the diagnosis, treatment, and prognosis of COVID-19. These co-infections have the potential to exacerbate disease symptoms and increase the mortality rate [3].

The incidence rate of influenza cases has gradually reduced during the period of the COVID-19 pandemic in 2021 [4]. Implementation of lockdown, other COVID-19 control measures and mobility restrictions probably had a role to play affecting prevalence of influenza and other respiratory viruses as well. Several analyses of surveillance reports stated that there has been a decrease in the circulation of influenza during the pandemic period [5, 6] Due to SARS CoV-2 infection, the influenza or other respiratory viruses infections were reduced among patients [7, 8]. The pattern of occurrence and prevalence other common

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respiratory viruses in the pandemic context is crucial in understanding the outcome of COVID-19 to prevent its widespread. In this study we have analyzed the prevalence of common respiratory viruses among COVID-19 positives and negative patients with either acute respiratory infections (ARI) or severe acute respiratory infections (SARI) cases referred to KIPMR, Chennai from the government hospitals of Tamil Nadu during the period from March 2020 to February 2022.

Methods

Case definition

Patients with all age groups were included in this study. The World Health Organization (WHO) defined SARI cases as people who were hospitalized within 10 days of the commencement of their illness and had a measured or reported fever and a cough. The definition of ARI cases was the hospitalization is necessary due to a cough or respiratory problems that started within the last 10 days of the illness. As per the WHO, case definition of ARI/SARI samples were collected and tested [9].

Sample collection

This study was a retrospective cross-sectional study which includes nasopharyngeal swabs samples routinely received in the Department of Virology, KIPMR, Chennai for COVID 19 testing. Samples were collected in accordance with Human ethical guidelines and regulations. Patients provided consent were enrolled in this study. In case of minor (children and adolescents) informed consent from a parent and/or legal guardian for study participation were obtained.

Extraction of RNA for detection of SARS CoV-2 and other respiratory viruses

The collected samples were processed and subjected for RNA extraction according to the standard protocol. The RNA was extracted using RNase Mini kit (Qiagen GmbH, Hilden, Germany) as per the manufacturer's instructions. The extracted RNA was subjected to the Real time reverse transcription PCR (rRT-PCR) for detection of SARS CoV-2 and Respiratory viruses duplex PCR. The rRT-PCR assay was performed with TaqPath™ COVID-19 combo kit for the detection of SARS CoV-2. Concurrently, all specimens were subjected to detection of other respiratory viruses like influenza A [A(H1N1) pdm09, A (H3N2)] (Flu A), influenza B [B/Yamagata and B/Victoria lineages] (Flu B) along with house-keeping RNase P gene, respiratory syncytial virus (RSV) A and B, human metapneumovirus (HMPV), human parainfluenza viruses (HPIV) 1, 2, 3 and 4, rhinoviruses and adenoviruses using rRT-PCR which performed in the applied biosystems Realtime PCR (Quant Studio 5) [10].

Meteorological data

During the study period meteorological particulars (average temperature—°C, rainfall—mm and humidity—percentage) were collected from Meteorological Centre, Nungambakkam, Chennai.

Data analysis

Statistical analyses were performed using GraphPad Prism 8.0.2 (San Diego, CA). The level of significance was set at $p < 0.05$. The Chi- square test and Fisher's exact test was used to find the relationship between variables associated with the co-infections of COVID-19. The positivity rate of influenza and other respiratory infections were calculated for SARI and ARI cases.

Results

A total of 1480 samples collected from the community and hospital settings during the period March 2020 to February 2022 were tested.

- Out of the total 1480 subjects, 620 (41.8%) had SARI and 860 (58.1%) had signs and symptoms of ARI (WHO case definition).
- Out of the 620 SARI subjects tested for SARS CoV-2, 198 (31.9%) were positive and 422 (68.06%) were negative for SARS CoV-2.
- Out of the 860 ARI subjects tested for SARS CoV-2, 352 (40.9%) were positive and 508 (59%) were negative for SARS CoV-2.
- The incidence of SARS CoV-2 amongst SARI and ARI was different which statistically significant with p value < 0.001 (Fig. 1, Fischer's exact test).
- It was noted that 550 (37.2%) of the total 1480 were SARS CoV-2 positive and 930 (62.8%) were negative for SARS CoV-2 and both the parameters were statistically significant which amongst ARI and SARI. Table 1 shows summary of SARI and ARI patients with co-existing respiratory cases among SARS CoV-2 and non-SARS CoV-2.

SARS CoV-2 co-existing with other respiratory viral co-infections

Among the 550 SARS CoV-2 positive cases, there were 7 (1.2%) positives for other respiratory viruses. Out of these,

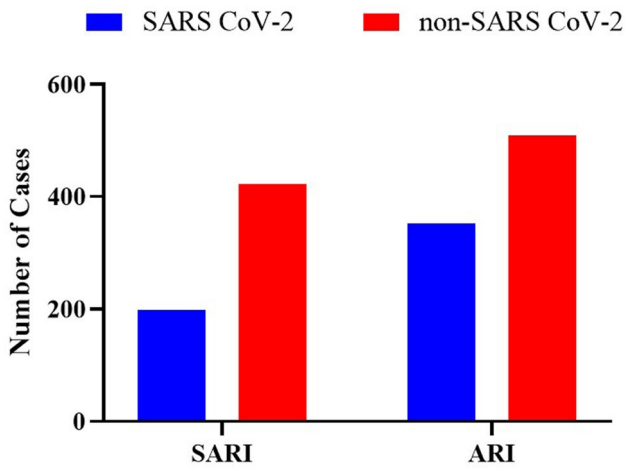


Fig. 1 Detection of SARS CoV-2 and non-SARS CoV-2 among SARI and ARI cases

Influenza A virus-H3N2 (Flu A) (n = 1), and Influenza B virus (Flu B) (n = 1) were found to be positive. We had 3 Respiratory Syncytial Virus (RSV) positives, of this 2 were RSV A and 1 was RSV B (Fig. 2). There were 2 Human Parainfluenza Virus 1 (HPIV 1) positives and we did not have any positivity for Adeno, Human Metapneumovirus (hMPV), Rhino and Boca virus. There were 7 viruses co-existing along with SARS CoV-2.

- Among the 550 positives, 198 were SARI subjects and 352 were ARI subjects.
- SARS CoV-2 positivity was 23.09% in the 31–55 years age group followed by 19–30 years (20.90%) age group.
- Co-infections with SARS CoV-2 occurred mainly in 19–30 years age group and in 13–18 years age groups; adult and geriatric populations showed less co-infection and there were no co-infections observed in pediatric population from age group 0–12 years.

Respiratory viruses among non-SARS CoV-2

We had 930 SARS CoV-2 negatives who were also tested for other respiratory viruses, out of this 222 (23.8%) were positive for other common respiratory viruses.

(a) Influenza viruses: We had got 82 (36.9%) positives for Influenza, of this 4 (0.4%) were H1N1 09, 40 (48.7%) were H3N2 positive and 38 (46.3%) were Flu B positives. Influenza virus positivity was 6.7% among ARI subjects with 3.9% positivity in SARI subjects in the SARS CoV-2 negative group.
(b) RSV: 71 (31.9%) were positive for RSV, of this 59 (83%) were RSV A positive and 12 (16.9%) were RSV B positive. RSV was higher in ARI group (7.6%) compared to SARI group (1%) among the SARS CoV-2 negatives.
(c) HPIV: We had 30 (13.5%) positives for HPIV, of this 17 (56%) were HPIV 1, 1 was HPIV 2, and 12 (40%) were HPIV 3.
(d) Adeno virus: 15 (6.7%) of them were positive for Adenoviruses.
(e) HMPV: There were 5 (2.2%) samples positive for human Metapneumovirus.
(f) Rhino virus: We had 17 (7.6%) samples positive for Rhino viruses.
(g) Boca virus: We had 2 samples positive for Boca virus.
(h) Co-infections: 13 (5.8%) samples had co infections among the respiratory viruses (Table 2).

Among the SARS CoV-2 negative cases (930), 222 (23.87%) were positive for other respiratory viruses. Highest positivity of RV (Respiratory Virus) was observed in the age group 6–12 years (40.18%) followed by 13–18 years (39.81%) and 0–5 years age groups (32.54%). Moderate positivity was noted in the age of 19–30 years (23.21%) and 31–55 years (16.02%), whereas less percentage of positivity was shown in geriatric population in the age of 56 years and

Table 1 Summary of SARI and ARI patients with co-existing respiratory cases among SARS CoV-2 and non-SARS CoV-2

Characteristics	Total cases	Influenza positives	RSV positives	HPIV positives	Adeno positives	hMPV positives	Rhino positives	Boca positives
SARI (n = 620)								
SARS CoV-2	198	1 (0.5%)	3 (1.5%)	1 (0.5%)	0	0	0	0
Non- SARS CoV-2	422	57 (13.5)	39 (9.2%)	21 (4.9%)	9 (2.1%)	3 (0.7%)	11 (2.6)	2 (0.47%)
ARI (n = 860)								
SARS CoV-2	352	1 (0.2%)	0	1 (0.2%)	0	0	0	0
Non- SARS CoV-2	508	25 (5%)	32 (6.2%)	9 (1.7%)	6 (1.1%)	2 (0.4%)	6 (1.1%)	0
Total	1480	84 (5.6%)	74 (5%)	32 (2.1%)	15 (1%)	5 (0.3%)	17 (1.1%)	2 (0.1%)

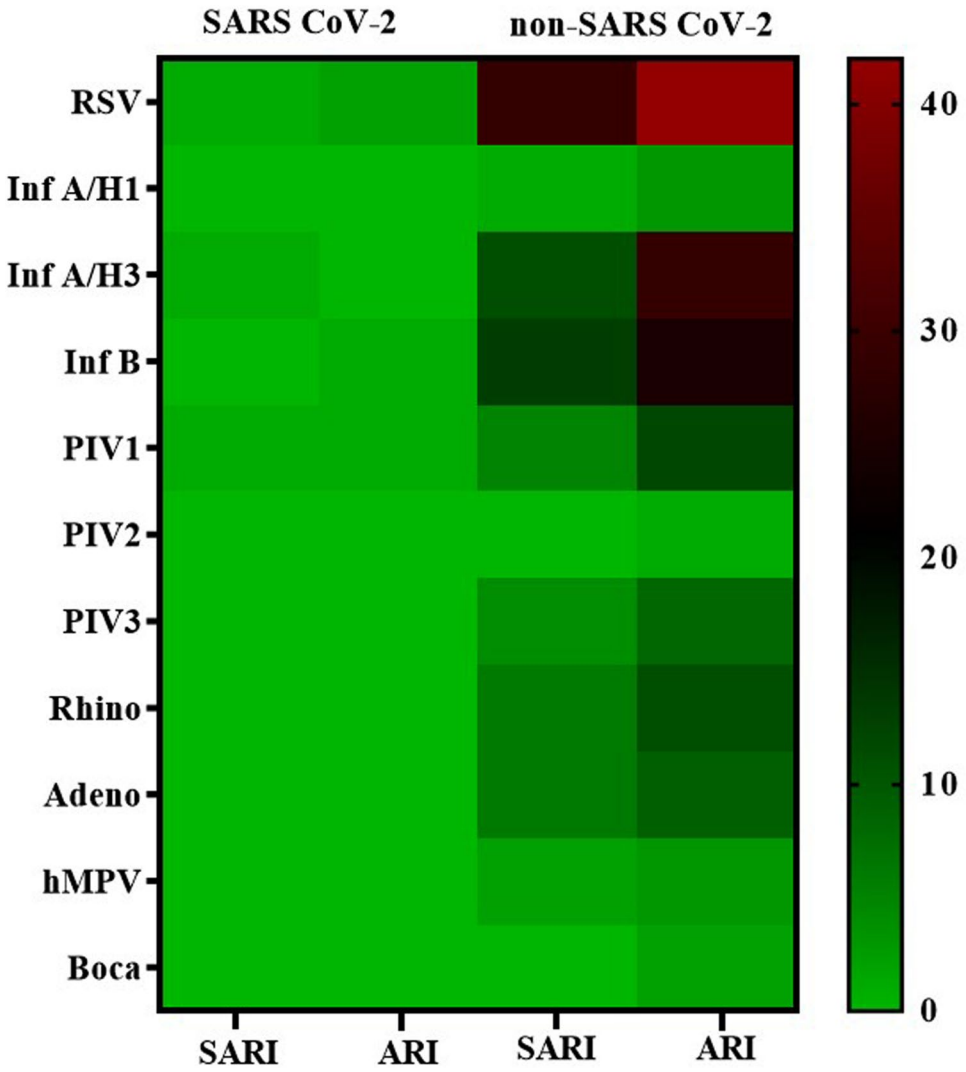


Fig. 2 Frequency heat-map of SARI and ARI cases with co-infections of respiratory viruses

above (9.19%). Importantly, pediatric age group (0–5 years) was typically affected by RSV (48%), followed by 22% of Influenza, 19.5% of Parainfluenza, 7% of Rhino and 2% of Adenovirus. In geriatric population, there was low prevalence of RV when compared to the other age groups, with influenza virus and RSV predominating (Table 3).

Role of weather conditions on occurrence of respiratory viruses

In this pandemic setting, it was observed that the respiratory infections were reported to be increased in the pre-monsoon and monsoon months of 2020. Mainly, influenza virus infections drastically increases in September with onset of monsoon and it reaches the peak in October and gradually reduces by the end of the year. The positivity of RV was 18% in pre-monsoon raised up to 42% in monsoon and reduces gradually. Significantly, the temperature and humidity correlated with increased RV activity during pre-monsoon and monsoon months (Fig. 3a and b). Besides, the detection of SARS CoV-2 was noted to be high in November and December months of 2020 and 2021. The backdrop witnessed consistent activity, with the COVID-19 positivity also displaying comparable peaks. (Fig. 4).

Discussion

We studied the prevalence of common respiratory viruses Influenza virus, RSV, Parainfluenza, Adeno infection in patients who are SARS CoV-2 positive and negative with clinical classification under the category of SARI and ARI respectively. In recent times, there are several reports studied the co-infections of SARS CoV-2 with other RV resulting in hospitalization and presenting with acute respiratory distress necessitating ICU care [11]. We had screened the subjects presenting with ARI and SARI for SARS CoV-2, then further both SARS CoV-2 negatives and positives tested for

Table 2 Distribution of Influenza and other Respiratory viruses among SARS CoV-2 and non-SARS CoV-2

COVID details	Total sample	Positives	Flu A H1	Flu A H3	Flu B	RSV A	RSV B	HPIV 1	HPIV 2	HPIV 3	HPIV 4	Adeno	hMPV	Rhino	Boca	Co- infection
Non-SARS CoV-2	930	222 (23.87%)	4	40	38	59	12	17	1	12	0	15	5	17	2	13
SARS CoV-2	550	7 (1.2%)	0	1	1	2	1	2	0	0	0	0	0	0	0	7
Total cases	1480	229	4	41	39	61	13	19	1	12	0	15	5	17	2	20

other respiratory viruses for better understanding of co-existence of other respiratory viruses. Among the SARS CoV-2 positives, we had only 7 positives for other RV. In general, the prevalence of other respiratory viruses was lower during the pandemic, this is in comparison with the positivity during the pre-covid years, could be because of proactive public health control measures implemented by the Government in containing the pandemic [12]. The enforcement of wearing mask, social distancing and handwashing in the control of pandemic could be very well used to contain other outbreaks in future [13].

The co-existence in general was markedly lower in the SARS CoV-2 positive group when compared to the negative group. The frequency of SARS CoV-2 co-infections with other respiratory viruses depends on seasonal and geographical variability [14]. The presence of influenza co-infection with SARS CoV-2 patients was 4.5% and 0.4% in Asia and America respectively reported through systematic review meta-analysis studies. Similarly, several reports have published about pooled prevalence of influenza co-infection with SARS CoV-2 as 0.8% (September 2020) and 19% (February 2021) [1]. About 92% of studies showed that influenza A, B and RSV were very common co-infecting viruses [15]. We have also had very low positivity of respiratory viruses with Influenza (0.3%), RSV (0.5%), HPIV (0.3%) among the SARS CoV-2 positives. This could be because of active multiplication of SARS CoV-2 would not have supported establishment of infection with other viruses. Other respiratory viruses, Influenza, RSV, HPIV, Rhino, Adenoviruses were higher in the ARI group than in the SARI group. Similarly, recent reports stated that other respiratory viruses were found as coinfection with SARS CoV-2 [16–18]. We had observed 13 coinfections among the respiratory viruses other than SARS CoV-2. We had seen only seven cases in which there was co-existence of SARS CoV-2 with other respiratory viruses. It was noted that the occurrence of other viral infections in the backdrop of pandemic were during the monsoon months of September till the cooler month of February indicating a positive seasonal pattern and prevalence of other respiratory viruses. ARI subjects had higher positivity than SARI subjects for SARS CoV-2 and for other respiratory viruses as well. The higher positivity rates in ARI group compared to SARI could be because most of the SARI cases are referred subjects and samples are probably not collected within 6 days for molecular analysis. Significantly, the incidence of SARI cases was lower co-infection with SARS CoV-2 during the pandemic wave of COVID-19 [19].

In general, respiratory viruses were seen predominantly in the above aged 30 year group implying that adults and those with underlying conditions are at increased risk towards developing respiratory infections. RSV was seen predominantly in the less than 5 year age group and 5–12 years age group and this is an established fact. Influenza was seen in

Table 3 Demographics and clinical characteristics of SARI and ARI cases with other respiratory viruses

Age (years)	Tested (n = 480)	Positives (n = 229)	%	Influenza (n = 84)	RSV (n = 74)	HPIV (n = 32)	Adeno (n = 15)	hMPV (n = 5)	Rhino (n = 17)	Boca (n = 2)
0–5	171	41	23.98	9	20	8	1	0	3	0
5–12	165	45	27.27	11	20	8	3	0	3	0
12–18	197	45	22.84	21	13	9	2	1	2	0
18–30	283	42	14.84	19	9	3	4	2	4	0
30–55	358	38	10.61	16	9	4	3	2	2	1
+ 55	306	18	5.88	8	3	0	2	0	3	1

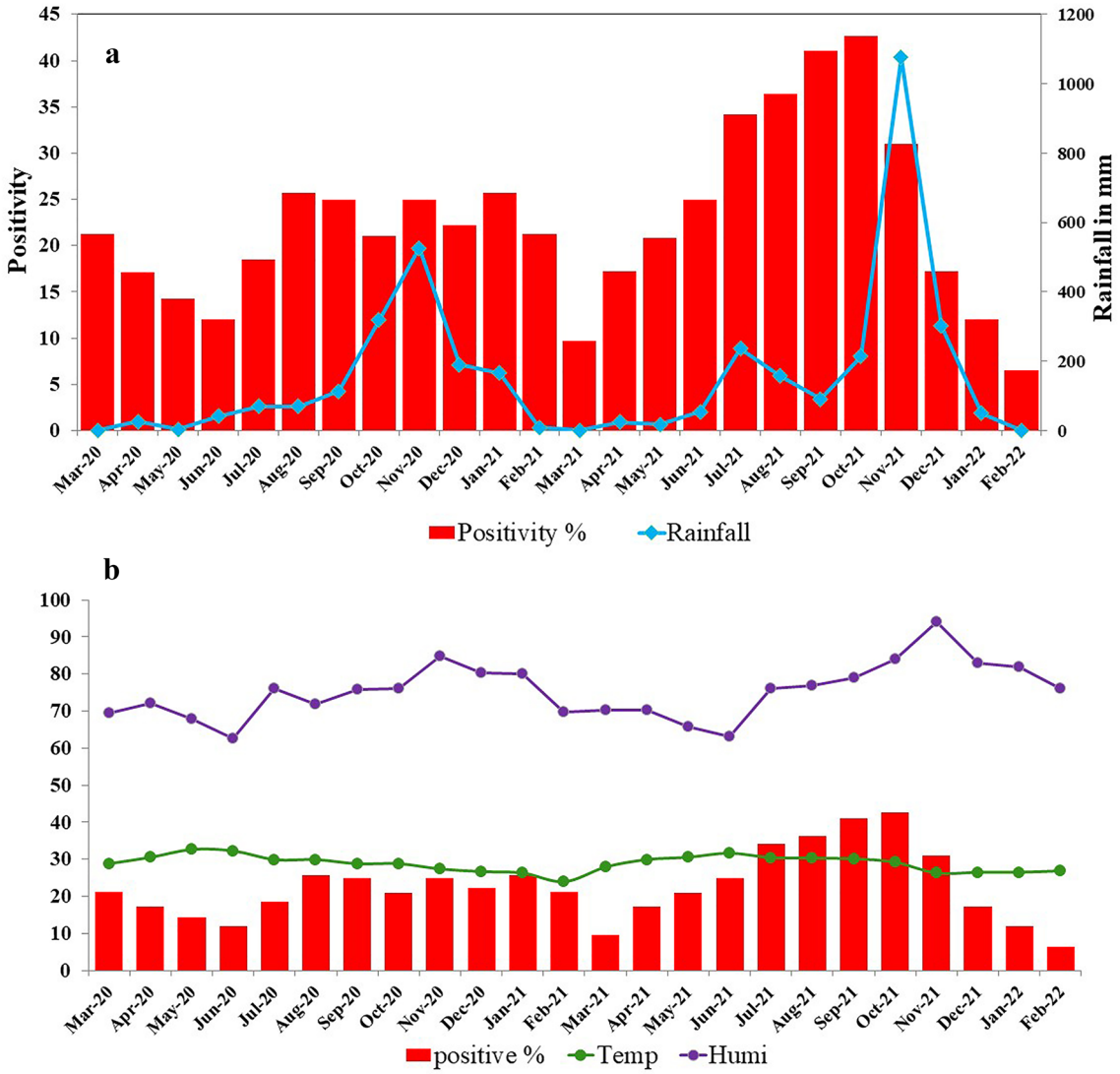


Fig. 3 **a** Time series of respiratory virus positivity with respect to rainfall. **b** The number of influenza cases detected month wise with respect to climate and temperature

the 12–18 year and 18–30 year age group predominantly. A recent study found that individuals between the ages of 19 and 60 had a higher risk of infecting both the influenza virus and SARS-CoV-2 [20]. Based on the results of this study, the existing review study reported that mean age of 52 years had SARS CoV-2 and influenza virus co-infections

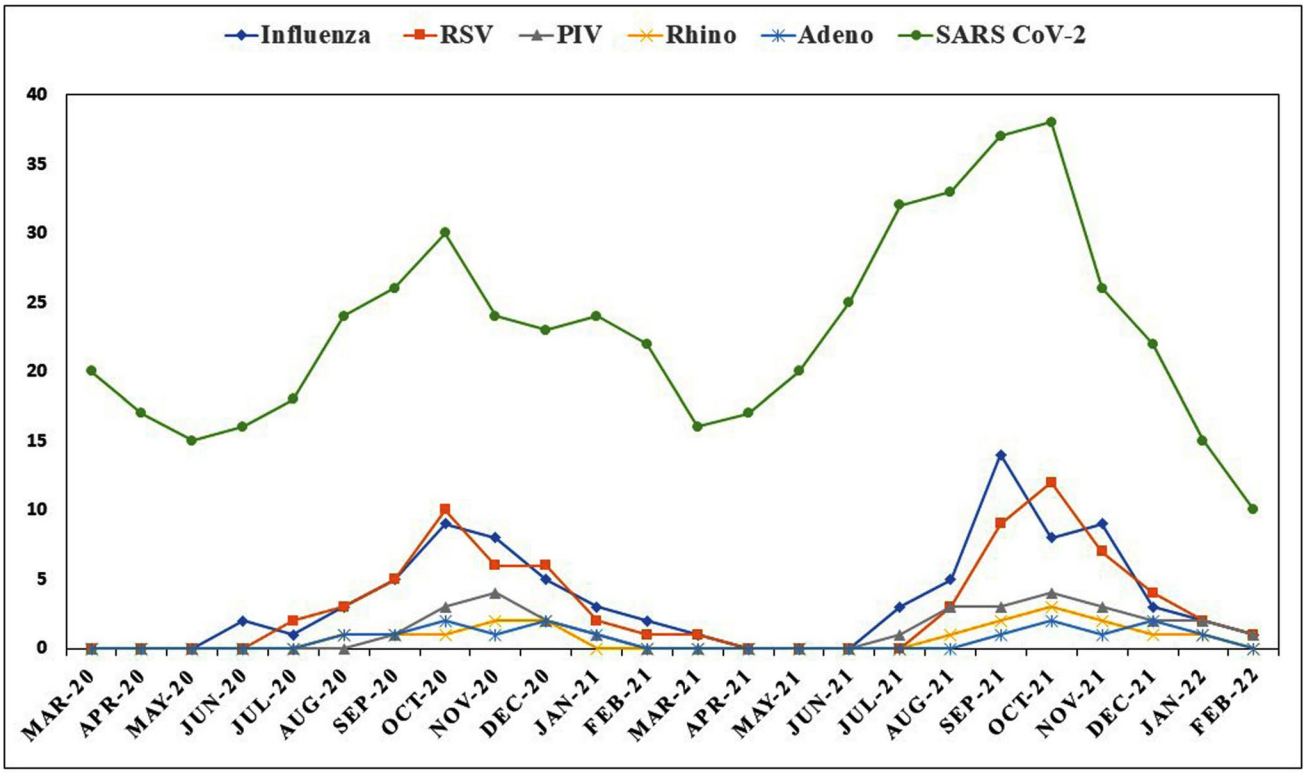


Fig. 4 Month-wise distribution of respiratory viruses in 2020–2022

that significantly border line [21]. On a common note, the immune systems of children and adults are different, both with respect to their physiological composition and immune responsiveness. Multiple hypothesis exists towards lesser impact of SARS CoV-2 on children when compared to adults like, presence of maternal antibodies, maturity of immune system etc. During the study period, with the pandemic predominating the entire period the occurrence of other respiratory viruses adhered to an established pattern of increase in the prevalence during pre-monsoon and monsoon seasons, with increase in rainfall and lower cooler climate. Globally, sentinel surveillance programs have been established at the national level to strengthen and understand the proportion of respiratory pathogens coexisting with SARS-CoV-2.

Conclusion

In this study, the occurrence of other respiratory diseases were reduced in COVID-19 positive cases and it very well matches with the seasonal climate pattern. The prevalence of respiratory viruses along with COVID-19 can alter the manifestation and prognosis of COVID-19. Hence parallel inspecting for other viral infections on a sentinel surveillance mode can aid the public health officials and physicians in assessing prognosis and management as well.

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Author contributions SM performed the experiments and drafted the manuscript. SM, and KM collected clinical specimens and information, MAK, CPA and KK performed the data analysis. KK and CPA participated in the study design and coordinated the drafting of the manuscript. All the authors read and approved the final manuscript.

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Data availability All data generated or analysed during this study are included in this manuscript.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval The study was carried out after obtaining ethical approval from Institutional Ethics Committee of King Institute of Preventive Medicine and Research, India (KIPMR/IEC/2020/565 dated 06/01/2020). The study was performed in compliance with the standard ethical guidelines.

Informed consent Yes.

Consent for publication Yes.

References

1. Dadashi M, Khaleghnejad S, Abedi Elkhichi P, Goudarzi M, Goudarzi H, Taghavi A et al (2021) COVID-19 and Influenza co-infection: a systematic review and meta-analysis. Front Med 8:1–9
2. Geng Y, Wang Y (2023) Stability and transmissibility of SARS-CoV-2 in the environment. J Med Virol 95(1):28103
3. Wiersinga WJ, Rhodes A, Cheng AC, Peacock SJ, Prescott HC (2020) Pathophysiology, transmission, diagnosis, and treatment of coronavirus disease 2019 (COVID-19): a review. JAMA 324:782
4. Jayaram A, Jagadesh A, Kumar AMV, Davtyan H, Thekkur P, Del Rio Vilas VJ et al (2022) Trends in influenza infections in three states of India from 2015–2021: has there been a change during COVID-19 pandemic? Trop Med Infect Dis. 7(6):110
5. Galli C, Pellegrinelli L, Bubba L, Primache V, Anselmi G, Delbue S et al (2021) When the COVID-19 pandemic surges during influenza season: Lessons learnt from the sentinel laboratory-based surveillance of influenza-like illness in lombardy during the 2019–2020 season. Viruses 13(4):695
6. Olsen SJ, Azziz-Baumgartner E, Budd AP, Brammer L, Sullivan S, Pineda RF et al (2020) Decreased influenza activity during the COVID-19 pandemic—United States, Australia, Chile, and South Africa, 2020. MMWR Morb Mortal Wkly Rep 69(37):1305
7. Parums DV (2021) Editorial: a decline in influenza during the COVID-19 pandemic and the emergence of potential epidemic and pandemic influenza viruses. Medi Sci Monit. 27:e934949
8. Agca H, Akalin H, Saglik I, Hacimustafaoglu M, Celebi S, Ener B (2021) Changing epidemiology of influenza and other respiratory viruses in the first year of COVID-19 pandemic. J Infect Public Health 14(9):1186
9. WHO (2013) Global epidemiological surveillance standards for influenza. WHO, Geneva
10. Chadha M, Prabhakaran AO, Choudhary ML, Biswas D, Koul P, Kaveri K et al (2022) Multisite surveillance for influenza and other respiratory viruses in India: 2016–2018. PLOS Glob Public Health 2(11):e0001001
11. Ma L, Wang W, Le GJM, Wang X, Du S, Li C et al (2020) Coinfection of SARS-CoV-2 and other respiratory pathogens. Infect Drug Resist. 13:3045
12. Principi N, Autore G, Ramundo G, Esposito S (2023) Epidemiology of respiratory infections during the COVID-19 pandemic. Viruses 15(5):1160
13. Chiu NC, Chi H, Tai YL, Peng CC, Tseng CY, Chen CC et al (2020) Impact of wearing masks, hand hygiene, and social distancing on influenza, Enterovirus, and all-cause pneumonia during the Coronavirus pandemic: retrospective national epidemiological surveillance study. J Med Internet Res 22(8):21257
14. Hazra A, Collison M, Pisano J, Kumar M, Oehler C, Ridgway JP (2020) Coinfections with SARS-CoV-2 and other respiratory pathogens. Infect Control Hosp Epidemiol 41(10):1228–1229
15. Musuuza JS, Watson L, Parmasad V, Putman-Buehler N, Christensen L, Safdar N (2021) Prevalence and outcomes of co-infection and superinfection with SARS-CoV-2 and other pathogens: a systematic review and metaanalysis. PLoS ONE 16:1–23
16. Gautam S, Malhotra B, Sharma P, Bhomia N, Deeba F, Sharma H et al (2023) Co-infection of respiratory pathogens in SARS-CoV-2 positive patients at North West India, Rajasthan. Indian J Med Microbiol 44:100371
17. Morales-Jadan D, Muslin C, Viteri-Dávila C, Coronel B, Castro-Rodríguez B, Vallejo-Janeta AP et al (2023) Coinfection of SARS-CoV-2 with other respiratory pathogens in outpatients from Ecuador. Front Public Health 11:1264632
18. Arguni E, Supriyati E, Hakim MS, Daniwijaya EW, Makrufardi F, Rahayu A et al (2022) Co-infection of SARS-CoV-2 with other viral respiratory pathogens in Yogyakarta, Indonesia: a cross-sectional study. Ann Med Surg 77:103676
19. Aggarwal N, Potdar V, Vijay N, Mukhopadhyay L, Borkakoty B, Manjusree S et al (2022) SARS-CoV-2 and influenza virus co-infection cases identified through ILI/SARI sentinel surveillance: a pan-India report. Viruses 14(3):627
20. Rezaee D, Bakhtiari S, Jalilian FA, Doosti-Irani A, Asadi FT, Ansari N (2023) Coinfection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and influenza virus during the COVID-19 pandemic. Arch Virol 168(2):53
21. Maltezou HC, Papanikolopoulou A, Vassiliu S, Theodoridou K, Nikolopoulou G, Sipsas NV (2023) COVID-19 and respiratory virus co-infections: a systematic review of the literature. Viruses 15(4):865

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